

CODEALPHA
PYTHON PROGRAMMING INTERNSHIP
PROJECT REPORT

TASK 2: STOCK PORTFOLIO TRACKER

A Graphical Stock Investment Tracker Using Python

Submitted By:

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Internship Domain:

Python Programming

Organization:

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Academic Year:

2026-2027

Project Type:

Internship Task – 2

STOCK PORTFOLIO TRACKER

DECLARATION

I hereby declare that the project entitled “**Stock Investment Tracker**” has been developed as part of the **CodeAlpha Python Programming Internship – Task 2**.

This project was developed using Python programming concepts and provides a graphical interface for selecting stocks, entering quantities, calculating investment values, and displaying the total portfolio investment.

The stock prices used in the application are predefined for educational demonstration. The screenshots included in this report are the actual outputs of the developed application.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to CodeAlpha for providing me with the opportunity to participate in the Python Programming Internship.

I am thankful for the practical tasks provided during the internship, which helped me apply Python programming concepts to develop a useful graphical application.

I would also like to thank my teachers, mentors, friends, and family for their guidance, encouragement, and support throughout the development of this project.

The Stock Investment Tracker helped me improve my understanding of Python dictionaries, variables, user input, conditional statements, loops, arithmetic operations, and graphical user interface development.

Finally, completing this project provided practical experience in designing an application, implementing calculations, validating inputs, and presenting portfolio information clearly.

ABSTRACT

The Stock Investment Tracker is a Python-based graphical application developed as part of the CodeAlpha Python Programming Internship – Task 2.

The main purpose of the project is to calculate and display the total investment value of selected stocks using manually defined stock prices. The application provides a user-friendly dashboard where users can select a stock, enter the quantity, and add the investment to a portfolio.

The application uses a predefined dictionary containing stock symbols and their corresponding prices. When a stock and quantity are selected, the program calculates the investment value using basic arithmetic and updates the total investment.

The project demonstrates fundamental Python concepts including dictionaries, variables, input handling, conditional statements, loops, arithmetic operations, GUI components, and table-based data presentation.

The screenshots in this report demonstrate the actual working application, beginning with an empty portfolio and progressing through AAPL, GOOGL, and MSFT investments.

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1. INTRODUCTION

A stock portfolio consists of different stocks owned by an individual or organization. Tracking the value of these investments manually can become difficult when multiple stocks are involved.

The Stock Investment Tracker is a simple Python GUI application developed to calculate the total investment value based on predefined stock prices and quantities entered by the user.

The project uses a dictionary to store stock symbols and their corresponding prices. The user selects a stock and enters the number of shares. The application calculates the investment value of each entry by multiplying its price by the quantity.

The values of all entered investments are added together to determine the total investment value. The result is displayed prominently at the top of the dashboard, while individual investments are listed in the portfolio table.

2. PROBLEM STATEMENT

The objective of this project is to develop a simple Stock Investment Tracker using Python with a graphical user interface.

The application should allow users to select stock names and quantities, retrieve predefined prices, calculate individual investment values, and display the total investment value.

3. OBJECTIVES

1. To develop a simple stock investment calculator using Python.
2. To store stock prices using a Python dictionary.
3. To provide a graphical user interface for stock selection.
4. To allow users to enter the quantity of stocks.
5. To calculate the investment value of each stock.
6. To calculate and display the total investment value.
7. To support multiple portfolio entries.
8. To display portfolio details in a structured table.
9. To demonstrate Python programming concepts through a practical project.
10. To provide a foundation for future portfolio-management features.

4. SCOPE OF THE PROJECT

The scope of this project is to create a simple portfolio tracking application for calculating the value of selected stocks. The application uses manually defined stock prices stored in a Python dictionary.

The current version does not retrieve live stock prices from the internet or use an external financial API. It is intended for educational purposes and demonstrates programming and GUI concepts rather than real-time financial market information.

The project can be extended in the future to include live market data, graphical analysis, portfolio history, database storage, and advanced financial features.

5. TECHNOLOGIES AND TOOLS USED

Python: Main programming language used to implement the application.

Python Dictionary: Used to store stock symbols and predefined prices.

GUI Components: Used for stock selection, quantity input, buttons, labels, and portfolio presentation.

Table/Tree View: Used to display stock, quantity, price, and investment values.

6. PROJECT DESCRIPTION

The Stock Investment Tracker provides a dashboard containing four major areas: Total Investment, Available Stocks, Add Investment, and Your Portfolio.

The Available Stocks section shows AAPL, TSLA, GOOGL, AMZN, and MSFT. In the Add Investment section, the user selects a stock and enters its quantity. The Add Stock button triggers the calculation.

The Portfolio section displays each added investment using four columns: Stock, Quantity, Price, and Investment. The total investment value is continuously updated whenever a new stock entry is added.

8. FEATURES OF THE APPLICATION

8.1 Predefined Stock Prices

The application contains a dictionary with manually defined stock prices.

8.2 Stock Selection

The user can select a stock from the available stock list.

8.3 Quantity Input

The user can enter the number of shares to be added.

8.4 Investment Calculation

The value of each stock is calculated by multiplying its price by the quantity.

8.5 Total Investment

The application calculates and displays the total value of all entered stocks.

8.6 Multiple Investments

The user can add multiple stocks and multiple entries to the portfolio.

8.7 Portfolio Table

The portfolio table presents the stock symbol, quantity, price, and investment value in an organized format.

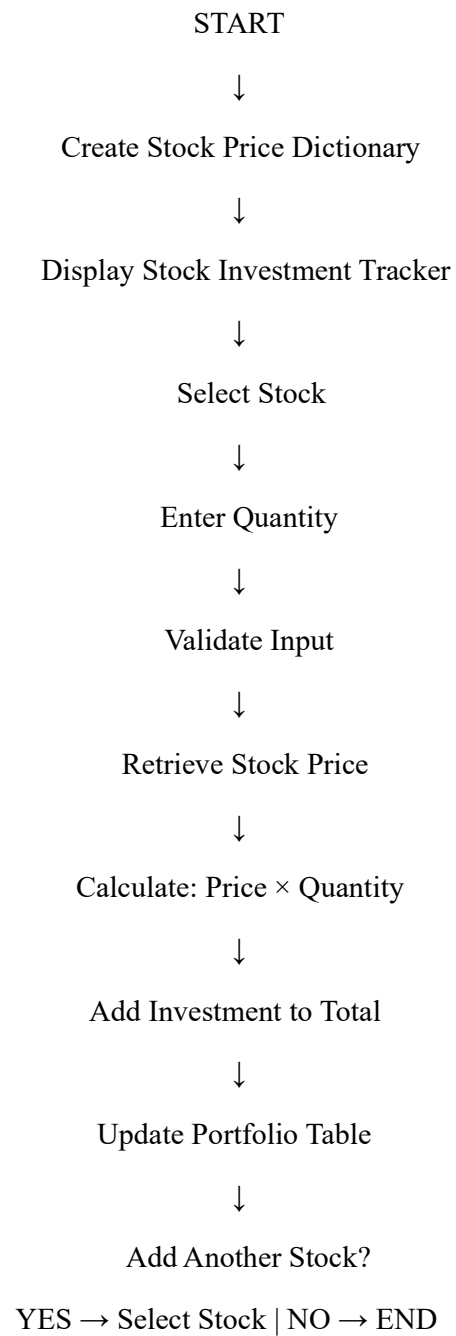
8.8 Simple User Interface

The dark-themed dashboard provides clearly separated sections and an easy-to-use input area.

9. METHODOLOGY

1. Define the project requirements.
2. Create a dictionary containing predefined stock symbols and prices.
3. Design the graphical user interface.
4. Allow the user to select a stock.
5. Accept the quantity entered by the user.
6. Validate the entered quantity and stock.
7. Retrieve the predefined stock price.
8. Calculate price $\times$ quantity.
9. Add the result to the total investment
10. Insert the stock information into the portfolio table

10.SYSTEM WORKFLOW



11. ALGORITHM

1. Start the program.
2. Create a dictionary containing predefined stock prices.
3. Initialize the total investment value to zero.
4. Display the graphical interface.
5. Display the available stocks.
6. Ask the user to select a stock.
7. Ask the user to enter the quantity.
8. Validate the input.
9. Retrieve the stock price from the dictionary.
10. Multiply the stock price by the quantity.
11. Add the calculated value to the total investment.
12. Add the investment details to the portfolio table.
13. Update the total investment display.
14. Repeat the process for additional stocks.
15. End the program.

12. PYTHON CONCEPTS USED

12.1 Dictionary

A Python dictionary is used to store stock symbols and their corresponding predefined prices.

Example: {"AAPL": 180, "TSLA": 250, "GOOGL": 150, "AMZN": 180, "MSFT": 420}

12.2 Variables

Variables are used to store stock names, quantities, prices, individual investment values, and the total investment.

12.3 Conditional Statements

Conditional statements can be used to validate the stock selection and quantity.

12.4 Loops

Loops can be used to support repeated entry of stock investments.

12.5 Arithmetic Operations

Multiplication is used for individual investment calculations and addition is used for the total portfolio value.

13. CALCULATION METHOD

The Stock Investment Tracker uses a simple mathematical calculation to determine the value of each investment.

$$\text{Investment Value} = \text{Stock Price} \times \text{Quantity}$$

Example 1 – AAPL

Price = \$180; Quantity = 50

$$\text{Investment Value} = 180 \times 50 = \$9,000$$

Example 2 – GOOGL

Price = \$150; Quantity = 34

$$\text{Investment Value} = 150 \times 34 = \$5,100$$

Example 3 – MSFT

Price = \$420; Quantity = 5

$$\text{Investment Value} = 420 \times 5 = \$2,100$$

Final additional MSFT entry

Price = \$420; Quantity = 23

$$\text{Investment Value} = 420 \times 23 = \$9,660$$

The final total shown in the supplied screenshot is:

$$\mathbf{\$9,000 + \$5,100 + \$2,100 + \$9,660 = \$25,860}$$

14. IMPLEMENTATION

The Stock Investment Tracker was implemented using Python and a graphical interface. A dictionary is created to store stock symbols and prices. GUI widgets are used for stock selection, quantity input, button interaction, and portfolio display.

When the user clicks Add Stock, the selected stock and quantity are read from the interface. The corresponding predefined price is retrieved, the investment value is calculated, and a new row is added to the portfolio. The total investment label is then updated.

15. TESTING AND RESULTS

Testing was performed to verify that the Stock Investment Tracker produces the expected results shown in the supplied screenshots.

Test Case	Input / Action	Expected Result	Result
1	Start application	Application opens successfully	Pass
2	Initial screen	Total Investment displays \$0	Pass
3	AAPL, Quantity 50	Investment = \$9,000	Pass
4	GOOGL, Quantity 34	Total = \$14,100	Pass
5	MSFT, Quantity 5	Total = \$16,200	Pass
6	MSFT, Quantity 23	Total = \$25,860	Pass
7	Multiple portfolio entries	Entries appear in table	Pass

16.OUTPUT SCREENSHOTS

Figure 1: Stock Investment Tracker – Starting Screen

The starting screen shows the Stock Investment Tracker dashboard. The Total Investment is initially \$0, the available stocks are displayed, and the portfolio table is empty.

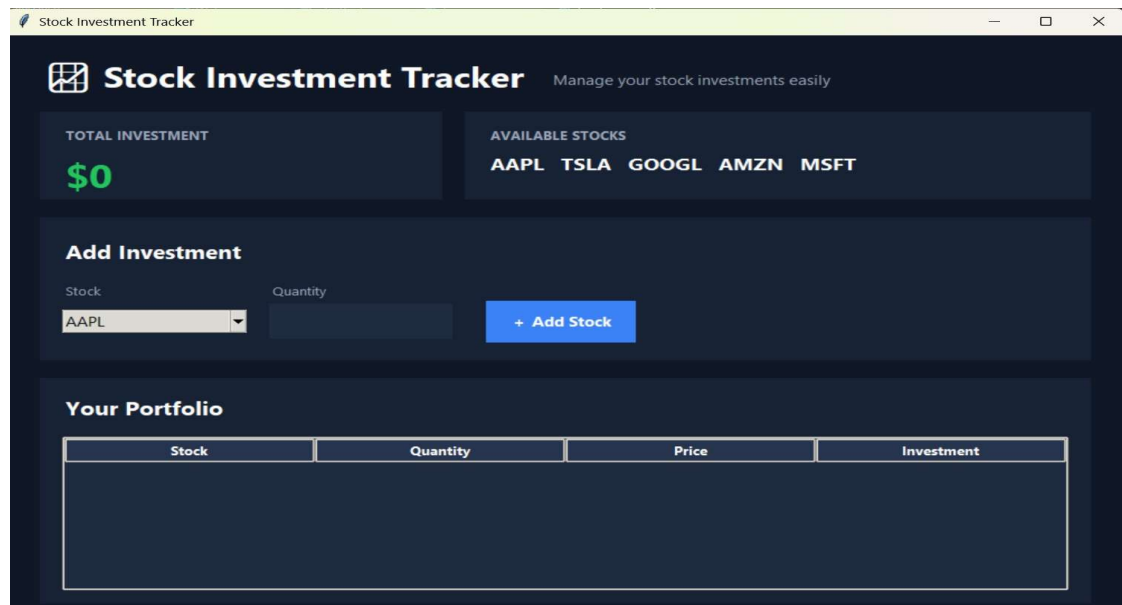


Figure 2: AAPL Investment Added

The user adds 50 shares of AAPL at the predefined price of \$180. The application calculates an investment of \$9,000 and updates the Total Investment.

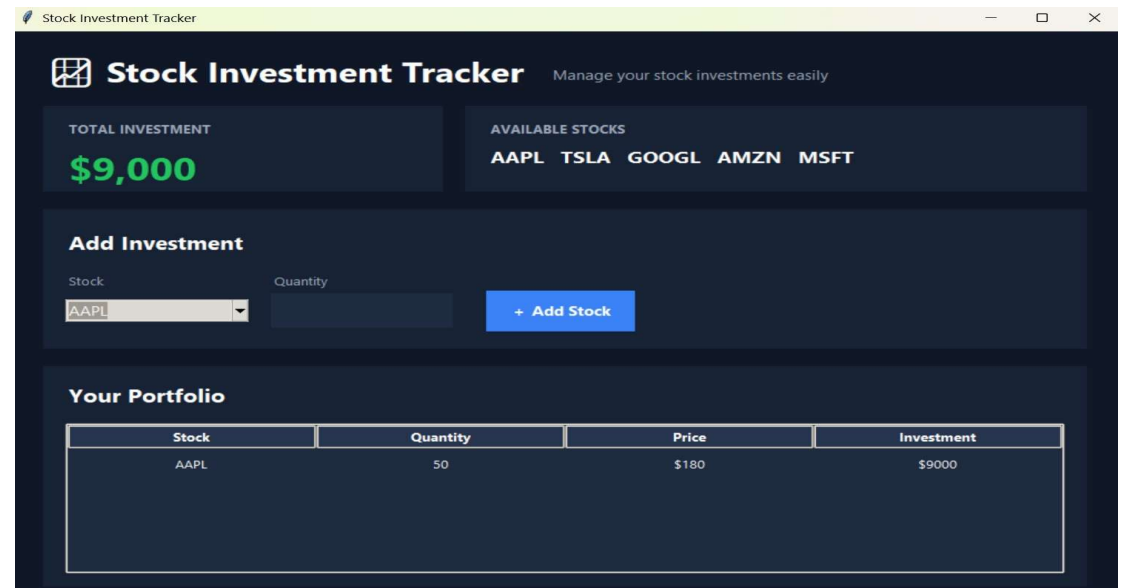


Figure 3: GOOGL Investment Added

The user adds 34 shares of GOOGL at the predefined price of \$150. The investment is \$5,100 and the running total becomes \$14,100.

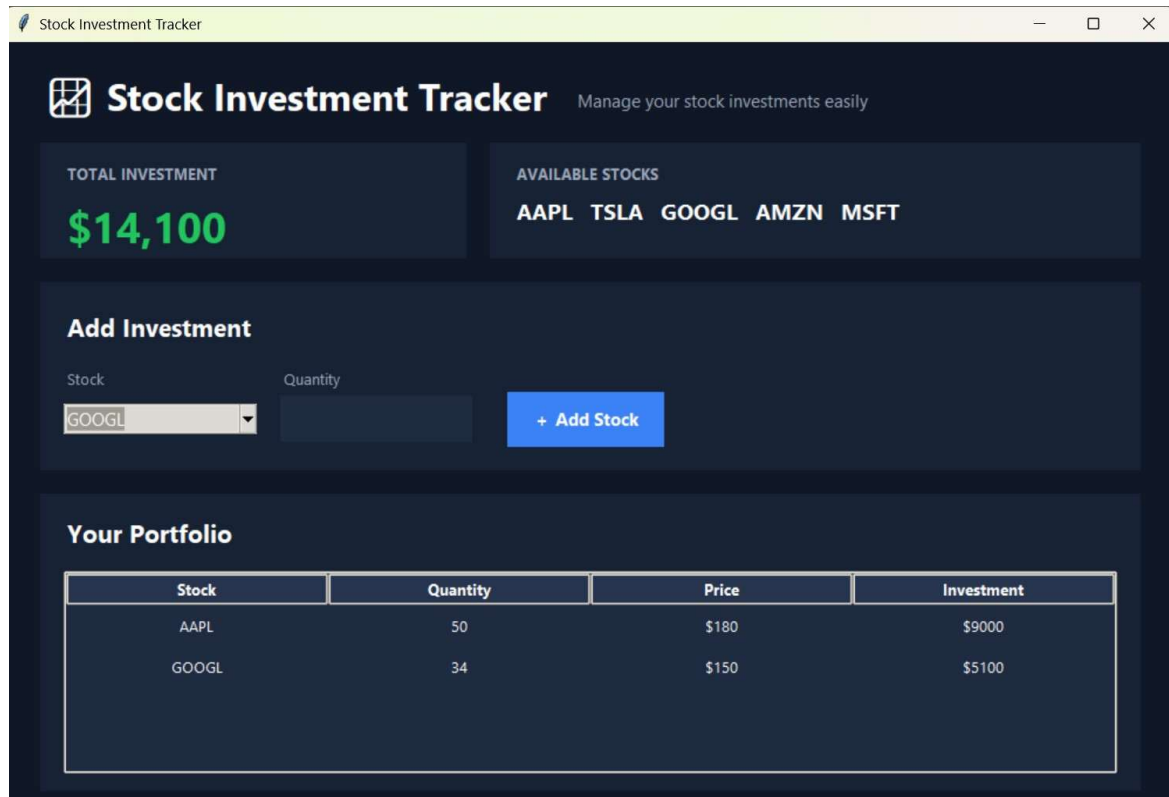


Figure 4: MSFT Investment Added

The user adds 5 shares of MSFT at the predefined price of \$420. The investment is \$2,100 and the running total becomes \$16,200.

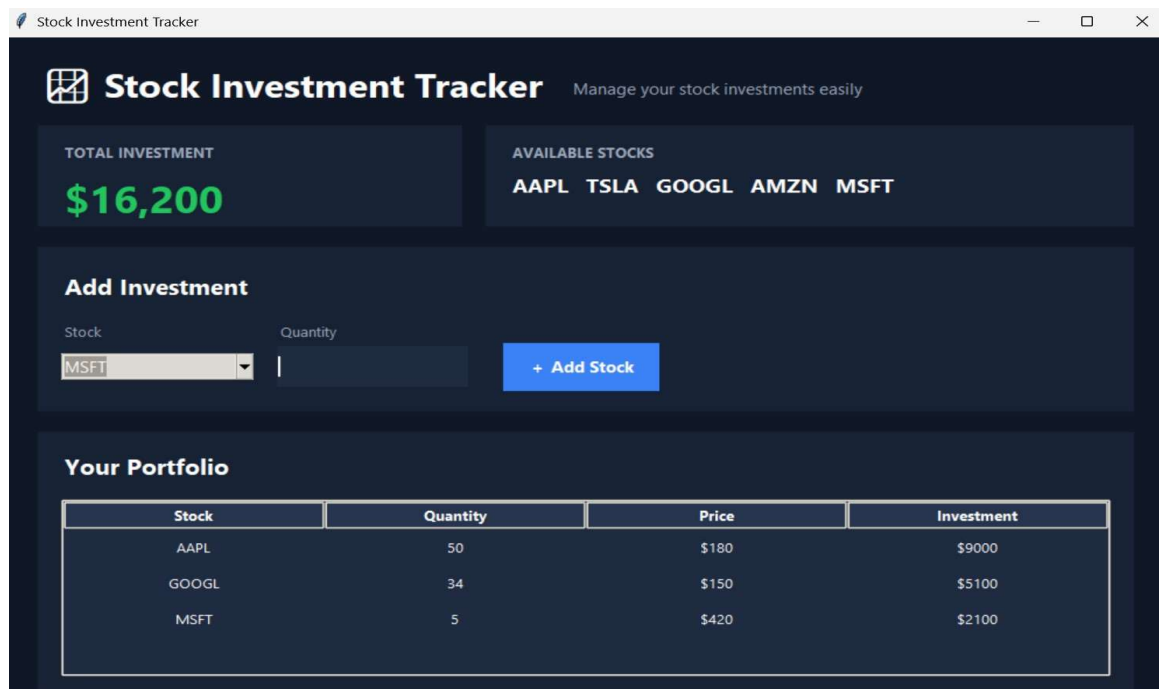
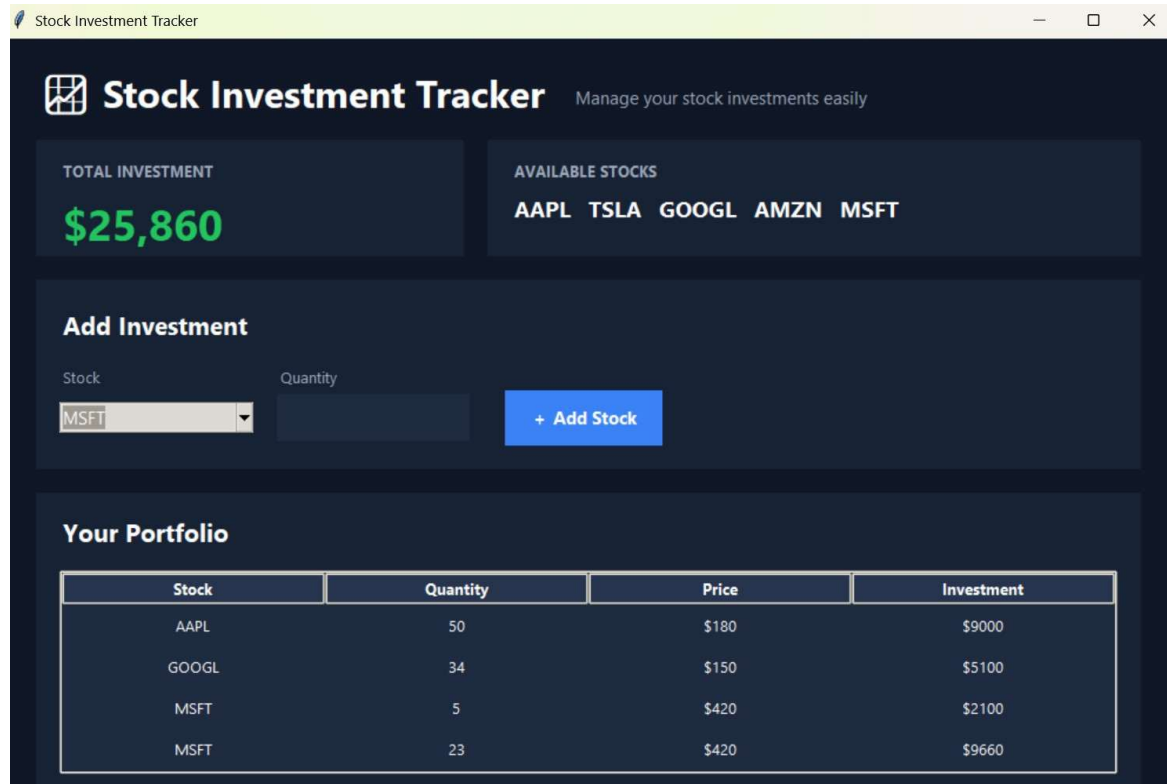


Figure 5: Final Portfolio

The final screenshot shows an additional MSFT entry of 23 shares at \$420, worth \$9,660. The displayed Total Investment is \$25,860.



17.ADVANTAGES

- Simple and easy to use.
- Easy to understand for beginners.
- Demonstrates practical use of Python dictionaries.
- Performs investment calculations quickly.
- Supports multiple stock entries.
- Displays the total investment clearly.
- Does not require an internet connection for predefined prices.
- Does not require a database for the basic version.
- Provides a structured portfolio table.
- Can be extended with additional features.

18. LIMITATIONS

1. Stock prices are manually defined.
2. The application does not provide real-time stock prices.
3. Prices are not automatically updated according to the stock market.
4. The project does not connect to a financial API.
5. The application provides basic portfolio calculations only.
6. Advanced financial analysis is not included.
7. The current version does not provide graphical investment charts.
8. Portfolio data is not permanently stored unless file/database support is added.

19. FUTURE ENHANCEMENTS

19.1 Real-Time Stock Prices

An external stock market API could be integrated to retrieve current stock prices.

19.2 Database Integration

A database could be used to permanently store portfolio information.

19.3 Portfolio Charts

Charts and graphs could be added to visually represent the distribution of investments.

19.4 Profit and Loss

The application could calculate the difference between purchase price and current market price.

19.5 Multiple Portfolio Support

Users could create and manage multiple portfolios.

19.6 Export Options

Portfolio information could be exported to CSV, Excel, or PDF formats.

19.7 Transaction History

The application could maintain a history of stock purchases and sales.

20. CONCLUSION

The Stock Investment Tracker was successfully developed using Python as a graphical stock investment calculation application.

The project successfully implements the main requirements by using predefined stock prices and allowing the user to select stock names and quantities. The application calculates individual investment values using basic arithmetic and calculates the total investment value of the portfolio.

The supplied screenshots demonstrate the complete working process. The starting screen displays a total investment of \$0. After adding AAPL, GOOGL, and MSFT entries, the total is updated step by step to \$9,000, \$14,100, \$16,200, and finally \$25,860 after the additional MSFT investment.

Through this project, practical knowledge of Python dictionaries, variables, input handling, conditional statements, loops, arithmetic operations, GUI development, and tabular data presentation was gained.

Although the current application uses manually defined prices, it provides a strong foundation for developing a more advanced stock portfolio management system with live market data, database storage, charts, profit/loss analysis, export options, and transaction history.